

Senior High School

General Biology I

Quarter II – Module 4

Calvin Cycle



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SELF-LEARNING MODULE

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Introductory Message

For the facilitator:

Welcome to the General Biology I 12 Self-Learning Module on Calvin Cycle.

This module was collaboratively designed, developed and reviewed by educators both from public and private institutions to assist you, the teacher or facilitator in helping the learners meet the standards set by the K to 12 Curriculum while overcoming their personal, social, and economic constraints in schooling.

This learning resource hopes to engage the learners into guided and independent learning activities at their own pace and time. Furthermore, this also aims to help learners acquire the needed 21st century skills while taking into consideration their needs and circumstances.

In addition to the material in the main text, you will also see this box in the body of the module:



Notes to the Teacher

This contains helpful tips or strategies that will help you in guiding the learners.

As a facilitator, you are expected to orient the learners on how to use this module. You also need to keep track of the learners' progress while allowing them to manage their own learning. Furthermore, you are expected to encourage and assist the learners as they do the tasks included in the module.

For the learner:

Welcome to the General Biology I 12 Self-Learning Module on Light Reactions.

This module was designed to provide you with fun and meaningful opportunities for guided and independent learning at your own pace and time. You will be enabled to process the contents of the learning resource while being an active learner.

CONTENT STANDARD

The learners demonstrate understanding of Respiration.

PERFORMANCE STANDARD

The learner should be able to prepare simple fermentation setup using common fruits to produce wine or vinegar via microorganisms.

LEARNING COMPETENCY

- Describe the significant events of the Calvin cycle
(STEM_BIO11/12-IIa-j-5)

LEARNING OBJECTIVES

At the end of the module, you should be able to:

1. Describe the sequence of events in the light-independent photosynthetic reactions;
2. Realize the importance of the sun in sustaining life processes; and
3. Make an illustration that will connect the Calvin cycle to the light-dependent reaction.

INTRODUCTION

You, like all organisms on Earth, are a carbon-based life form. In other words, the complex molecules of your amazing body are built on carbon backbones. You might already know that you're carbon-based, but have you ever wondered where all of that carbon comes from?

As it turns out, the atoms of carbon in your body were once part of carbon dioxide (CO₂) molecules in the air. Carbon atoms end up in you, and in other life forms, and this is the second stage of photosynthesis, known as the Calvin cycle (or the light independent reactions).

REVIEW OF THE PREVIOUS MODULE

The goal of the light-dependent stage of photosynthesis is to convert light energy from the sun to chemical energy. But what is the overall outcome of the light reactions in photosynthesis?

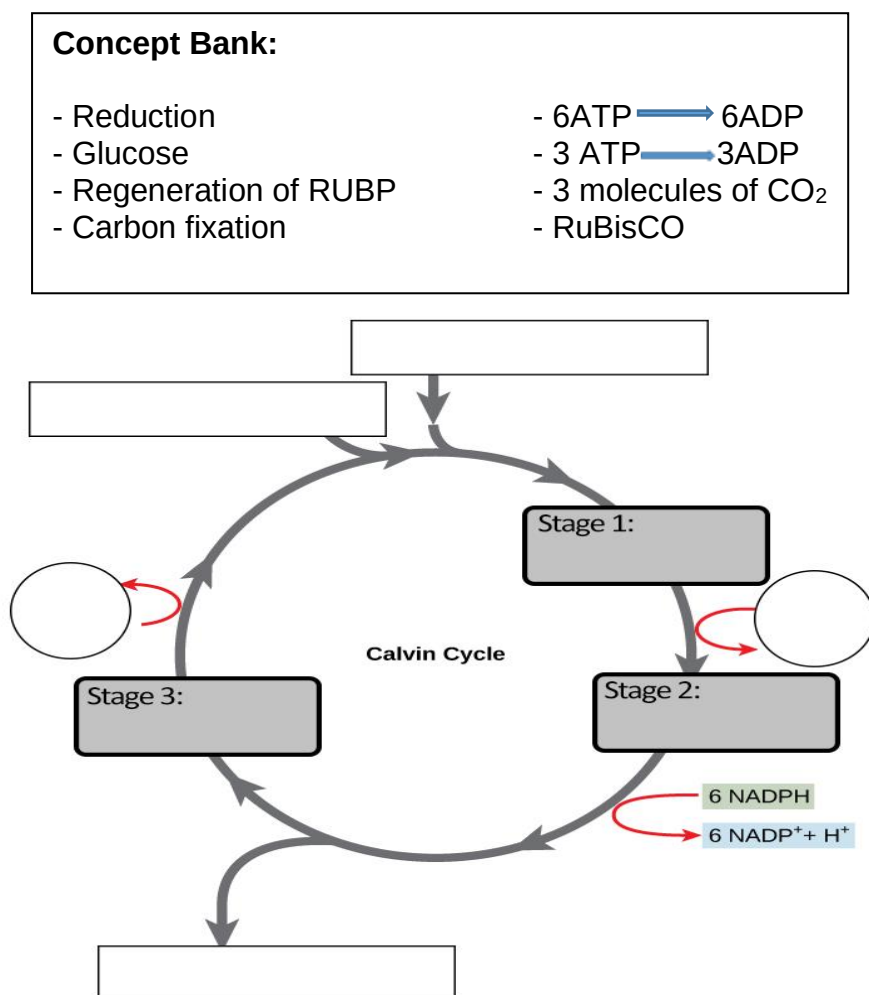
PRESENTATION OF THE NEW MODULE

The cell has the fuel required to create food in the form of carbohydrate molecules after the energy from the sun is transformed and packed into ATP and NADPH. The molecules of carbohydrates produced will have a backbone of carbon atoms. Where is this carbon coming from? Carbon dioxide, the gas that animals exhale with each breath, comes from the carbon atoms used to create carbohydrate molecules. The Calvin cycle is the term used for photosynthesis reactions that use energy stored to form glucose and other carbohydrate molecules through light-dependent reactions.

ACTIVITY

“Label Me”

Directions: Fill in the missing word/term to complete the Calvin cycle. You may refer to the box below.



Source: <https://www.biologycorner.com/wp-content/uploads/calvin-cycle-label.png>

ANALYSIS

Directions: Answer the following questions based on your understanding.

1. Why is the third stage of Calvin cycle called the regeneration stage?

2. Why does it take three turns of the Calvin cycle to produce G3P, the initial product of photosynthesis?

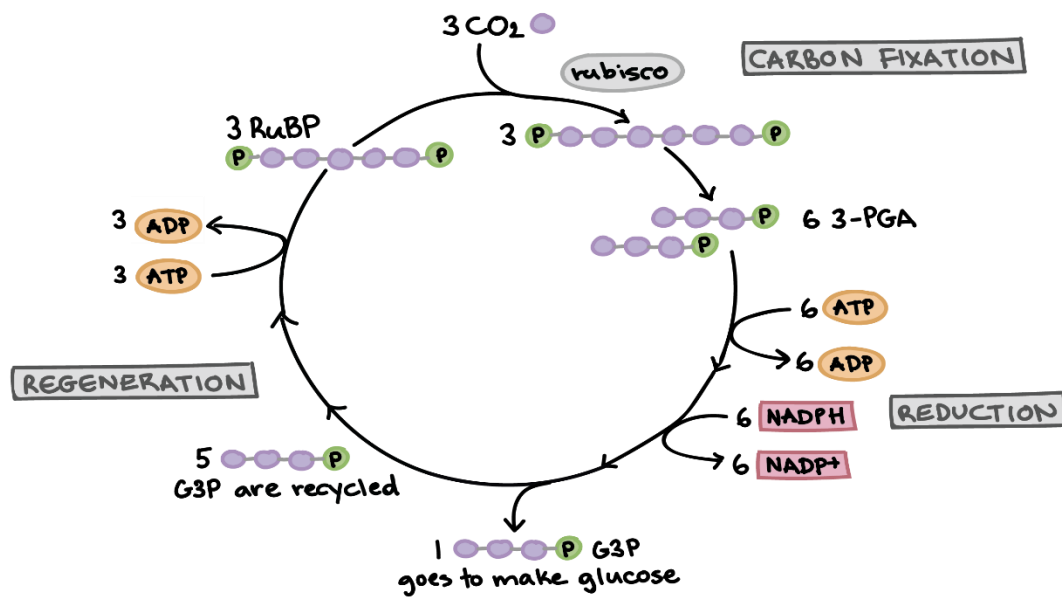
3. How does Calvin cycle convert CO₂ to glucose?

ABSTRACTION

Light-Independent Stage

The light-independent stage is also known as the Calvin cycle or Calvin-Benson cycle. It consists of reactions that use the energy stored and carried in the light-dependent-dependent reaction to build glucose. Carbon dioxide enters the chloroplast of plant cells through the stomata of the leaves. Inside the chloroplast, carbon dioxide diffuses into the stroma. An enzyme called ribulose – 1, 5 - bisphosphate carboxylase (RuBisCO) catalyzes the reaction and combines carbon dioxide with a five-carbon sugar molecule, which is called ribulose- 1, 5 - bisphosphate (RuBP). This results in an unstable six-carbon compound, which is immediately broken down into three-carbon compounds. This process is called carbon fixation because RuBP plucks carbon atoms from the carbon dioxide in the air attaches it to an organic molecule. The organic molecule is further modified into a simple sugar G3P using the energy from ATP and NADPH.

Some molecules of G3P are combined to form six carbon sugars, such as glucose (C₆H₁₂O₆). The remaining G3P molecules are used to regenerate the original organic molecule with energy from ATP. In summary, it will take six cycles of Calvin cycle to fix six carbon atoms from carbon dioxide.



Source: <https://www.khanacademy.org/science/ap-biology/cellular-energetics/photosynthesis/a/calvin-cycle>

Figure 1. Calvin cycle is also known as the dark reaction because it does not require the presence of light to produce glucose

Exercise 1

Directions: Match the letter of the item below to its most appropriate definition.

CHOICES:

- | | |
|-------------------------------|------------|
| A. Rubisco | E. Enzymes |
| B. RuBp | F. NADPH |
| C. Fructose 1, 6 Bisphosphate | G. ATP |
| D. CO ₂ | H. Sugar |

- _____ 1. What is eventually made by the Calvin cycle?
- _____ 2. The enzyme that fixes CO₂ to RuBp?
- _____ 3. The product formed when CO₂ and RuBp are fixed to one another.
- _____ 4. The atmospheric component of the Calvin cycle.
- _____ 5. This chemical bring phosphate from the photosystems to the Calvin cycle.

EXERCISE 2

Make an illustration of the Calvin cycle that will connect to the light-dependent reaction. Write your answer on the space provided.

APPLICATION

A graphic organizer is a visual representation used to organized information. To apply your learning in this lesson, make a graphic organizer summarizing the molecules involved on the Calvin cycle and the description of the three main stages.

Stages	Description	Molecules Involved

POST-TEST

Directions: Read the questions carefully and identify what is asked. Write the letter of the correct answer.

1. From which component of the light-dependent reactions does NADPH form most directly?
 - A. photosystem I
 - B. photosystem II
 - C. ATP synthase
 - D. cytochrome complex
2. The term 'rubisco' is frequently used in connection with photosynthesis. Rubisco is:
 - A. A set of criteria
 - B. A breakfast cereal
 - C. A guide for assigning score
 - D. An important enzyme in the Calvin cycle
3. Which of the following explains what happens to oxygen produced by the light-dependent reactions?
 - A. It is used in Calvin cycle
 - B. It is released into the atmosphere
 - C. It combines with NADPH to produce water.
 - D. It is recycled as a reactant in another light –dependent reactions.
4. A mutation occurs in the chloroplast of a lettuce plant causing its thylakoid membranes to become more permeable to charged ions. How might this mutation affect the Calvin cycle?
 - A. There would be no change to the Calvin cycle.
 - B. More carbohydrates would be produced by the Calvin cycle.
 - C. Fewer carbohydrates would be produced by the Calvin cycle.
 - D. none of the above
5. Some of the reactions of the Calvin cycle are similar to reactions of the:
 - A. Glycolysis
 - B. Fatty acid synthesis
 - C. Tricarboxylic acid cycle
 - D. Pentose phosphate pathway

6. The photosynthetic dark reactions.
- A. occur only in the dark
 - B. convert carbon dioxide to carbohydrates
 - C. produced the reduced coenzyme NADH
 - D. are the main source of ATP in photosynthesis
7. Which is the correct order of the stages of the Calvin cycle?
- A. Carboxylation---Regeneration---Reduction
 - B. Regeneration---Carboxylation---Reduction
 - C. Reduction---Carboxylation---Regeneration
 - D. Carboxylation---Reduction---Regeneration
8. In the Calvin cycle, more ATP than NADPH is used, how is this difference made up?
- A. the cyclic pathway creates more ATP
 - B. additional ATP is created from glucose
 - C. excess NADPH is reused in the light reaction
 - D. the Calvin cycle occurs half as often as the light dependent reaction.
9. Which of the following likely explains why?
- A. Sunlight is important in activating carbon fixation in the Calvin cycle.
 - B. The enzyme involved in the Calvin cycle are unable to bind substrates in the dark.
 - C. The Calvin cycle requires ATP and NADPH, which requires sunlight to be produced.
 - D. RuBP regeneration requires sunlight in order to occur and continue the Calvin cycle.
10. Because the reactions of the Calvin cycle can occur in darkness, this portion of photosynthesis is often referred to as the dark reactions. Although they can occur in darkness, in reality, they take place most often in periods of sunlight because.
- A. The enzyme cannot find each other in the dark.
 - B. Sunlight activates the enzymes of the Calvin cycle.
 - C. Dark reactions are too sinister for life forms on this planet.
 - D. The reactions of the Calvin cycle require ATP and NADPH, necessitating sunlight for generation.
11. Which complex is not involved in the establishment of conditions for ATP synthesis?
- A. ATP synthase
 - B. photosystem I
 - C. photosystem II
 - D. cytochrome complex

12. Which of the following structures is not a component of a photosystem?
- A. ATP synthase
 - B. reaction center
 - C. antenna molecule
 - D. primary electron acceptor
13. The light-dependent reactions occur in the __, while the Calvin cycle occurs in the__.
- A. stroma; nucleus
 - B. stroma; thylakoid membranes
 - C. thylakoid membranes; stroma
 - D. cytoplasm;; thylakoid membrane
14. Which molecule must enter the Calvin cycle continually for the light-independent reactions to take place?
- A. CO₂
 - B. RuBP
 - C. 3-PGA
 - D. RuBisCO
15. What are the products of light dependent reaction accustomed power the light-independent reaction?
- A. Sugar and ATP
 - B. NADPH and ATP
 - C. Oxygen and NADPH
 - D. NADPH and carbon acid gas

REFLECTION

You did it well! Congratulations on finishing the supplementary learning module! You have just had an amazing learning journey and for sure, you will also do the same in the succeeding modules.

For the last time, share your final insight by completing the following sentence prompts. It was such a wonderful learning experience with you.

3	Things I learned today...
2	Things I found interesting...
1	1 Question I still have...

GLOSSARY

The following terms used in this module are defined as follows for you to be guided.

Adenosine triphosphate also known as ATP, is a molecule that carries energy within cell

Calvin cycle also known as light-independent stage; consists of reactions that use energy stored and carried in the light-dependent reaction to build glucose.

Carbon fixation the process of converting inorganic CO₂ gas into organic compounds.

Enzyme biological molecules (typically proteins) that significantly speed up the rate of virtually all of the chemical reactions that take place within cells.

Stomata tiny openings or pores in plant tissue that allow for gas exchange.

Stroma a thick fluid space in the inner membrane of the chloroplast.

Thylakoid a small disc-shaped and interconnected membranous sacs suspended within stroma.

ANSWER KEY

POST-TEST

1. A
2. D
3. B
4. C
5. D
6. B
7. D
8. A
9. C
10. D
11. B
12. A
13. C
14. A
15. B

ANALYSIS

- Answer may vary

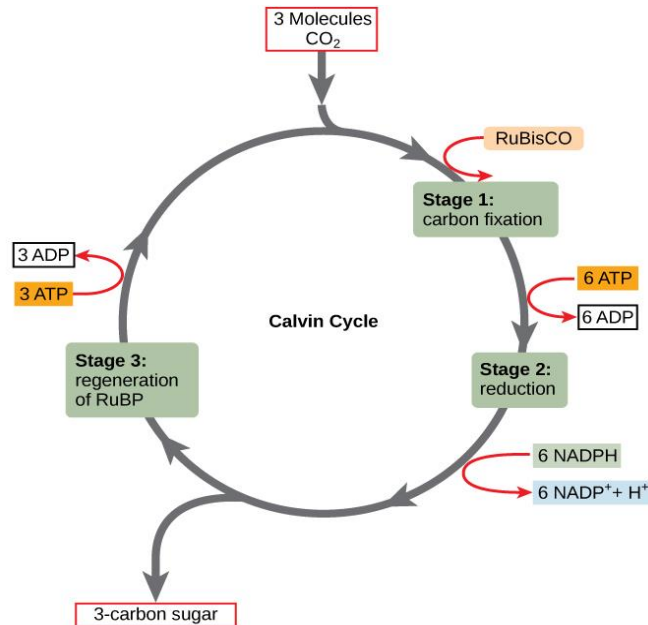
EXERCISE 1

1. H
2. A
3. C
4. D
5. G

REVIEW OF THE PREVIOUS MODULE

-The production of ATP and NADPH

ACTIVITY



APPLICATION

Stage	Description
Carbon fixation	A CO ₂ molecule combines with a five-carbon acceptor molecule, ribulose-1,5-bisphosphate (RuBP). This step make a six-carbon compound that splits into two molecules of a three-carbon compound, 3-phosphoglyceric acid (3-PGA). This reaction is catalyzed by the enzyme RuBP carboxylase or rubisco.
Reduction	ATP and NADPH are used to convert the 3-PGA molecules of a three-carbon sugar, glyceraldehyde-3-phosphate (G3P). This stage gets its name because NADPH donates electrons to, or reduces, a three-carbon intermediate to make G3P
Regeneration	Some G3P molecules go to make glucose, while others must be recycled to regenerate the RuBP acceptor. Regeneration requires ATP and involves a complex network of reactions.

REFERENCES

Morales-Ramos, A.C; and Ramos, J. D; 2017. Exploring Life Through Science Series General Biology 1. Phoenix Publishing House.

Basco-Tiamzon, M.A and Rabago, L.M. 2016. General Biology 1 Teacher's Manual. Published by Vibal Group, Inc.

Realuyo, Z.M; and Layug-Martinez, E.A; 2015. Biological Science. Published by Mindshapers Co., Inc.

Online Resources

- <https://www.khanacademy.org/science/biology/photosynthesis-in-plants/the-calvin-cycle-reactions/a/calvin-cycle>
- <https://pressbooks.bccampus.ca/conceptsofbiologymolnarcamosun/chapter/5-3-the-calvin-cycle/>
- <https://www.biologycorner.com/worksheets/calvin-cycle.html>

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